André Panisson, Ph.D.

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PROFESSIONAL SUMMARY

Machine Learning Researcher and AI Systems Architect with 15+ years spanning academic innovation and industrial R&D. Specialized in graph neural networks, responsible AI, and scalable social data systems. Proven track record of building explainable ML models, real-time data pipelines, and advancing scientific understanding of complex systems. Published in top-tier venues including WWW, KDD, AALT, xAI, IJCNN, and EPJ. Currently leading AI research at CENTAI Institute.

CORE TECHNICAL EXPERTISE

- **Responsible AI**: Explainability (XAI), Fairness, Transparency, Human-centric AI Systems
- **Generative AI**: Large Language Models (LLMs), Document Analysis, Retrieval-Augmented Generation (RAG)
- **AI for Risk & Compliance**: Anti-financial crime, Regulatory alignment, Behavioral pattern detection
- **Machine Learning**: Graph Neural Networks, Adversarial Learning, Model Evaluation
- **Systems & Engineering**: Real-time data pipelines, Distributed systems, Simulation of social-technical systems
- **Programming**: Python, Java, C/C++, PL/SQL, JavaScript, Bash
- **Libraries & Tools**: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, D3.js, Spark, Hadoop
- **Domains**: Computational social science, Generative AI, Misinformation detection, Network science

PROFESSIONAL EXPERIENCE

CENTAI Institute — *Principal Researcher, Team Lead – Responsible AI*
Turin, Italy | 2022–Present

- Lead the **Responsible AI team**, designing methodologies and tools that promote **explainability, fairness, and transparency** in AI systems, with a focus on accountability and human-centric design.
- Directed **industry collaborations** applying **generative AI** to streamline legal document analysis, improving efficiency and interpretability.
- Oversaw the development of AI solutions to **enhance anti-financial crime compliance** in banking environments, aligning technical innovation with regulatory compliance.
- Led applied research projects on **detection of online coordinated inauthentic behavior**, contributing to safer digital ecosystems.
- Mentored junior researchers and guided interdisciplinary initiatives blending AI, transparency and policy within complex socio-technical systems.

ISI Foundation — *Senior Research Scientist*
Turin, Italy | 2012–2022

- Engineered data analysis tools and ML models to explore human behavior, epidemic spreading, and digital trace analytics
- Built real-time interaction monitoring systems with wearable sensors (SocioPatterns)
- Delivered open-source tools for visual analytics (e.g., datainterfaces.org, ebolatracking.org)
- Oversaw collaborations on AI fairness and healthcare analytics

Insight Data Science — *Fellow*
Palo Alto, CA | 2012

- Created a real-time hot-topic tracker on Twitter using graph-based user expansion (Python + D3.js)

University of Turin — *PhD Researcher & Lecturer*
Turin, Italy | 2009–2011

- Developed recommender systems and mobile-aware collaborative filtering algorithms
- Created the Gephi Graph Streaming plugin for dynamic graph visualization
- Lectured in cryptography, complex networks, and recommender systems

Xeffe / ValueTeam — *Software Architect*
Turin, Italy | 2008–2011

- Led architectural design for banking platforms (Java, Oracle)
- Optimized data pipelines and transaction throughput across legacy + modern systems

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EDUCATION

Ph.D., Computer Science
University of Turin, Italy | 2009–2011
Thesis: *Selective Information Dissemination for Mobile Computing*

M.Sc., Computer Science
Federal University of Rio Grande do Sul, Brazil | 2004–2006
Thesis: *P2P-Based Network Management Solution*

B.Sc., Computer Science
Federal University of Rio Grande do Sul, Brazil | 1999–2003

LANGUAGES

- Portuguese: Native | English: Full Professional | Italian: Full Professional

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TEACHING & TRAINING

- **Data Mining: Statistical Modeling and Learning from Data,** 2013 – 2024, Department of Physics, University of Turin, Italy.
- **Data Mining: Statistical Modeling and Learning from Data**, Computer Science Department of ENS Lyon, January 2016, Lyon, France
- **Big Dive** – a training program focused on Data Science, Development and Visualization, Oct 2012, June 2013, June 2014, June 2015 and June 2016, Turin, Italy
- **Fundamentals of Cryptography and Network Security** – Laboratory, University of Turin, VII Bando AA 2010/2011 art.33
- **Complex Neworks** – lessons about Graph Visualization, Graph Layouts and Dynamic Networks, Univ. of Turin
- **Computer Security II** – Laboratory, University of Turin, VIII Bando AA 2011/2012 art.33

SELECTED PUBLICATIONS

- **Fast and Effective GNN Training through Sequences of Random Path Graphs**, KDD 2025
- **Adversarial Online Collaborative Filtering**, ALT 2024
- **Evaluating Link Prediction Explanations for GNNs**, xAI 2023
- **Global Misinformation Spillovers in the Vaccination Debate**, JMIR 2023

- **Predicting City Poverty Using Satellite Imagery**, CVPRW 2019

PUBLICATIONS

F. Bonchi, C. Gentile, F.P. Nerini, A. Panisson, F. Vitale. **Fast and Effective GNN Training through Sequences of Random Path Graphs.** In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (KDD 2025).

Perotti, A., Borile, C., Miola, A., Nerini, F.P., Baracco, P. and Panisson, A. **Explainability, Quantified: Benchmarking XAI Techniques.** In *World Conference on Explainable Artificial Intelligence* (xAI 2024).

Piaggesi, Simone, Megha Khosla, André Panisson, and Avishek Anand. **Dine: Dimensional interpretability of node embeddings.** *IEEE Transactions on Knowledge and Data Engineering* (TKDE 2024)

Nerini, Francesco Paolo, Paolo Bajardi, and André Panisson. **Value is in the Eye of the Beholder: A Framework for an Equitable Graph Data Evaluation.** In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency* (FACCT 2024)

Monti, Corrado, Paolo Bajardi, Francesco Bonchi, André Panisson, and Alan Perotti. **A True-to-the-model Axiomatic Benchmark for Graph-based Explainers.** *Transactions on Machine Learning Research* (TMLR 2024).

S. Pasteris, F. Vitale, M. Herbster, C. Gentile, A. Panisson. **Adversarial Online Collaborative Filtering.** In Proceedings of the 35th International Conference on Algorithmic Learning Theory (ALT 2024).

A. Battiston, L. Napoli, P. Bajardi, A. Panisson, A. Perotti, M. Szell, R. Schifanella. **Revealing the determinants of gender inequality in urban cycling with large-scale data.** EPJ Data Science. 12(1): 9 (2023)

C. Borile, A. Perotti, A. Panisson. **Evaluating Link Prediction Explanations for Graph Neural Networks.** In: Explainable Artificial Intelligence. xAI 2023. Communications in Computer and Information Science, vol 1902. (2023)

F. Dibitonto, F. Garcea, A. Panisson, A. Perotti, L. Morra, L. **HOLMES: HOLonym-MEronym Based Semantic Inspection for Convolutional Image Classifiers.** In: Explainable Artificial Intelligence. xAI 2023. Communications in Computer and Information Science, vol 1902. (2023)

A. Perotti, S. Bertolotto, E. Pastor, A. Panisson. **Beyond One-Hot-Encoding: Injecting Semantics to Drive Image Classifiers.** In: Explainable Artificial Intelligence. xAI 2023. Communications in Computer and Information Science, vol 1902, 525-548. (2023)

A. Perotti, P. Bajardi, F. Bonchi, A. Panisson. **Explaining Identity-aware Graph Classifiers through the Language of Motifs.** Proceedings of the 2023 International Joint Conference on Neural Networks (IJCNN). (2023)

J. Lenti, Y. Mejova, K. Kalimeri, A. Panisson, D. Paolotti, M. Tizzani, M. Starnini. **Global Misinformation Spillovers in the Vaccination Debate Before and During the COVID-19 Pandemic: Multilingual Twitter Study.** JMIR Infodemiology. 2023 May 24;3:e44714. (2023)

A. Capozzi, G. D. F. Morales, Y Mejova, C Monti, and A. Panisson. **The Thin Ideology of Populist Advertising on Facebook during the 2019 EU Elections.** In Proceedings of the ACM Web Conference 2023 (WWW '23). (2023)

S. Piaggesi, A. Panisson, & G. Petri. **Effective Higher-order Link Prediction and Reconstruction from Simplicial Complex Embeddings.** In Learning on Graphs Conference (pp. 55-1). PMLR. (2022)

S. Piaggesi & A. Panisson. **Time-varying graph representation learning via higher-order skip-gram with negative sampling**. EPJ Data Science, 11(1), 33. (2022)

C. Rollo, G. De Francisci Morales, C. Monti, & A. Panisson. **Communities, Gateways, and Bridges: Measuring Attention Flow in the Reddit Political Sphere.** In Social Informatics: 13th International Conference, SocInfo 2022, Glasgow, UK, October 19–21, 2022, Proceedings (pp. 3-19). (2022)

G. Crupi, Y. Mejova, M. Tizzani, D. Paolotti, and A. Panisson. **Echoes through time: Evolution of the italian covid-19 vaccination debate.** In Proceedings of the International AAAI Conference on Web and Social Media (Vol. 16, pp. 102-113). (2022)

S. Giurgola, S. Piaggesi, M. Karsai, Y. Mejova, A. Panisson & M. Tizzoni. **Mapping urban socioeconomic inequalities in developing countries through facebook advertising data.** Frontiers in big Data, 5. (2021)

C. Panigutti, A. Perotti, A. Panisson, P. Bajardi, & D. Pedreschi. **FairLens: Auditing black-box clinical decision support systems**. Information Processing & Management, 58(5), 102657. (2021)

E. G. Amparore, F. Cinus, C. Maestri, L. Petrocchi, D. Polinelli, F. Scarpa, A. Perotti, A. Panisson, P. Bajardi. **Forecast of Distributed Energy Generation and Consumption in a Partially Observable Electrical Grid: A Machine Learning Approach**, 2021 IEEE Madrid PowerTech, pp. 1-6. (2021)

M. Starnini, C. E. Tsourakakis, M Zamanipour, A. Panisson, W. Allasia, M. Fornasiero, L. Li Puma, V. Ricci, S. Ronchiadin, A. Ugrinoska, M. Varetto, D. Moncalvo. **Smurf-Based Anti-money Laundering in Time-Evolving Transaction Networks.** Joint European Conference on Machine Learning and Knowledge Discovery in Databases. Springer, Cham. (2021)

A. Capozzi, G. De Francisci Morales, Y. Mejova, C. Monti, A. Panisson, & D. Paolotti. **Clandestino or Rifugiato? Anti-immigration Facebook Ad Targeting in Italy**. In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (pp. 1-15). (2021)

F. Poletto, Y. Zhang, A. Panisson, Y. Mejova, D. Paolotti, & S. Ponserre. **Developing Annotated Resources for Internal Displacement Monitoring**. In Companion Proceedings of the Web Conference 2021 (pp. 136-144). (2021)

A. Capozzi, G. De Francisci Morales, Y. Mejova, C. Monti, A. Panisson, & D. Paolotti. **Facebook Ads: Politics of Migration in Italy**. In International Conference on Social Informatics (pp. 43-57). (2020)

N. Gozzi, M. Tizzani, M. Starnini, F. Ciulla, D. Paolotti, A. Panisson, & N. Perra. **Collective response to media coverage of the COVID-19 pandemic on Reddit and Wikipedia: mixed-methods analysis.** Journal of medical Internet research, 22(10), e21597. (2020)

F. Cinus, F. Bonchi, C. Monti, A. Panisson. **Generating Realistic Interest-Driven Information Cascades.** In Proc. of the International AAAI Conference on Web and Social Media (Vol. 14, pp. 107-118). (2020)

M. Tizzoni, A. Panisson, D. Paolotti, C. Cattuto. **The impact of news exposure on collective attention in the United States during the 2016 Zika epidemic.** PLoS computational biology 16 (3), e1007633 (2020)

B. Edizel, F. Bonchi, S. Hajian, A. Panisson, T. Tassa. **FaiRecSys: Mitigating algorithmic bias in recommender systems.** International Journal of Data Science and Analytics, 9(2), 197-213. (2020)

D. Balsamo, P. Bajardi, A. Panisson. **Firsthand Opiates Abuse on Social Media: Monitoring Geospatial Patterns of Interest Through a Digital Cohort.** In Proceedings of the 2019 World Wide Web Conference. (2019)

S. Piaggesi, L. Gauvin, M. Tizzoni, C. Cattuto, N. Adler, S. Verhulst, A. Panisson. **Predicting City Poverty Using Satellite Imagery.** In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops (pp. 90-96). (2019)

B. Gobbo, D. Balsamo, M. Mauri, P. Bajardi, A. Panisson, P. Ciuccarelli. **Topic Tomographies (TopTom): a visual approach to distill information from media streams.** In Computer Graphics Forum (Vol. 38, No. 3, pp. 609-621). (2019)

L. Ozella, L. Gauvin, L. Carenzo, M. Quaggiotto, P.L. Ingrassia, M. Tizzoni, A. Panisson, D. Colombo, A. Sapienza, K. Kalimeri, F. Della Corte, C. Cattuto. **Wearable proximity sensors for monitoring a mass casualty incident exercise: a feasibility study** Journal of medical internet research 21 (4), e12251 (2019)

M.C. Kiti, M. Tizzoni, T.M. Kinyanjui, D.C. Koech, P.K. Munywoki, M. Meriac, L. Cappa, A. Panisson, A. Barrat, C. Cattuto, D.J. Nokes. **Quantifying social contacts in a household setting of rural Kenya using wearable proximity sensors.** EPJ Data Science 5.1: 1-21 (2016)

M.G. Beiró, A. Panisson, M. Tizzoni, C. Cattuto. **Predicting human mobility through the assimilation of social media traces into mobility models.** EPJ Data Science, 5:30 (2016)

A. Sapienza, A. Panisson, J. Wu, L. Gauvin, C. Cattuto. **Detecting anomalies in time-varying networks using tensor decomposition.** Proc. 5th IEEE ICDM Workshop on Data Mining in Networks (2015)

C. Levallois, M. Marchand, T. Mata, A. Panisson. **Twitter for Research Handbook**, EMLYON Press, 2015.

A. Sapienza, A. Panisson, J. Wu, L. Gauvin, C. Cattuto. **Anomaly Detection in Temporal Graph Data: An Iterative Tensor Decomposition and Masking Approach.** Proc. 1st International Workshop on Advanced Analytics and Learning on Temporal Data (AALTD 2015), (2015)

P. Bajardi, M. Delfino, A. Panisson, G. Petri, M. Tizzoni. **Unveiling patterns of international communities in a global city using mobile phone data.** EPJ Data Science 4, no. 1 (2015): 1-17.

L. Gauvin, A. Panisson, A. Barrat, C. Cattuto. **Revealing latent factors of temporal networks for mesoscale intervention in epidemic spread.** arXiv preprint arXiv:1501.02758(2015).

M. Génois, C.L. Vestergaard, J. Fournet, A. Panisson, I. Bonmarin, A. Barrat. **Data on face-to-face contacts in an office building suggest a low-cost vaccination strategy based on community linkers.** Network Science, 3(03), 326-347, (2015)

A. Panisson, L. Gauvin, M. Quaggiotto, C. Cattuto. **Mining Concurrent Topical Activity in Microblog Streams.** Proc. of the the 4th Workshop on Making Sense of Microposts, co-located with the 23rd Intl. World Wide Web Conference (WWW 2014), Seoul, Korea, April 2014 (Best Paper Award).

L. Gauvin, A. Panisson, C. Cattuto. **Detecting the community structure and activity patterns of temporal networks: a non-negative tensor factorization approach**. PLOS ONE 9.1 (2014): e86028.

L. Gauvin, A. Panisson, C. Cattuto, A. Barrat. **Activity clocks: spreading dynamics on temporal networks of human contact.** Scientific Reports, v. 3, p. 3099, 2013.

C. Cattuto, M. Quaggiotto, A. Panisson, A. Averbuch. **Time-varying social networks in a graph database.** In: First International Workshop on Graph Data Management Experiences and Systems - GRADES '13. New York: ACM Press, 2013.

A. Panisson, L. Gauvin, A. Barrat, C. Cattuto. **Fingerprinting temporal networks of close-range human proximity**. Intl. Workshop on the Impact of Human Mobility in Pervasive Systems and Applications (PerMoby 2013), March 18 2013, San Diego, CA

A. Panisson, A. Barrat, C. Cattuto, G. Ruffo, R. Schifanella. **On Human Proximity Dynamics for Data Diffusion through Ad-Hoc Networks**. Ad Hoc Networks, Special Issue on Social-Based routing in Mobile and Delay-Tolerant Netwoks. Volume 10, Issue 8, November 2012, Pages 1532-1543

C. Melchiors, D. T. Mattjie, C. R. P. dos Santos, A. Panisson, L. Z. Granville and L. M. Tarouco. **A P2P-Based Strongly Distributed Network Polling Solution.** Advancements in Distributed Computing and Internet Technologies: Trends and Issues. IGI Global, 2012. 289-313.

A. Basso, M. Milanesio, A. Panisson, G. Ruffo. **On Collaborative Filtering Techniques for Live TV and Radio Discovery and Recommendation**. In Proc. of the 12th International Conference on Electronic Commerce and Web Technologies (EC-Web 2011), Toulouse, France, August 29 - September 2, 2011.

A. Basso, M. Milanesio, A. Panisson. **From Recordings to Recommendations: Suggesting Live Events in the DVR Context.** In RecSys 2010: Proc. of the International Workshop on the Practical Use of Recommender Systems, Algorithms and Technologies, Barcelona, Spain, 30 September 2010.

Basso, M. Milanesio, A. Panisson**. Social Aspects of Video Recording.** In AISB10: Proc. of the Thirty Sixth Annual Convention of the Society for the Study of Artificial Intelligence and Simulation of Behaviour, Leicester, UK, 29 March - 1 April 2010.

C. Melchiors, A. H. dos Santos, D. Mattjie, C. R. dos Santos, A. Panisson, L. Z. Granville, L. M. R. Tarouco. **A network polling solution through a P2P-based distributed management environment.** In SAC '10: Proceedings of the 2010 ACM Symposium on Applied Computing (Sierre, Switzerland, March 22 - 26, 2010). ACM, New York, pp. 729-730.

R. Schifanella, A. Panisson, C. Gena and G. Ruffo. **MobHinter: Epidemic Collaborative Filtering and Self-Organization in Mobile Ad-Hoc Networks**. In RecSys 2008: Proceedings of the 2nd ACM Intern. Conf. on Recommender Systems, October 23-25, 2008, Lausanne, Switzerland. ACM Press.

C. C.. Marquezan, A. Panisson, L. Z. Granville, G. Nunzi, M. Brunner. **Maintenance of Monitoring Systems Throughout Self-Healing Mechanisms.** In DSOM 2008: Proceedings of the 19th IFIP/IEEE International Workshop on Distributed Systems: Operations and Management, 22-26 September 2008, Samos Island, Greece, pp. 176-188, 2008.

A. Panisson, G. Ruffo, and R. Schifanella. **X-hinter: a framework for implementing social oriented recommender systems.** In HT '08: Proceedings of the Nineteenth ACM Conference on Hypertext and Hypermedia (Pittsburgh, PA, USA, June 19 - 21, 2008). ACM, New York, NY, p. 235-236, 2008.

A. Panisson, D. M. Rosa, C. Melchiors, L.Z. Granville, M. J. B. Almeida, L. M. R. Tarouco. **Designing the Architecture of P2P-Based Network Management Systems.** In ISCC2006: Proceedings of the 2006 IEEE Symposium on Computers and Communications. p. 69-75, 2006.

L. Z. Granville, D. M. Rosa, A. Panisson, C. Melchiors, M. J. B. Almeida, L. M. R. Tarouco. **Managing Computer Networks Using Peer-to-Peer Technologies.** IEEE Communications Magazine, v. 43, n. 10, p. 62-68, 2005.

A. Panisson, M. J. B. Almeida, L. M. R. Tarouco, L. Z. Granville. **Implementação de um Algoritmo para Busca em Redes Peer-to-Peer. (Implementation of a Search Algorithm for P2P Networks).** In: Proc. of the Brazilian Workshop on Peer-to-Peer Systems (WP2P2005), 2005, Fortaleza. v. 1. p. 25-36.